

Probability Functions

Dr Lily Asquith (Lily, she/her)



Assignment Notes

Hopefully you have all decided on either the **Visualising Bayes** or **Explaining Monty Hall** options for the first assignment.

I finally found time to do the Visualizing Bayes option, and found that I could make the plot more informative by choosing different values from those specified on the assignment. It is fine if you go 'off piste' a little bit with those instructions, if you want to.

For the Monty Hall one-pager, I have uploaded an example latex template and it is also fine to use a google doc if latex is not useful for your future.

The Monty Hall Problem pollev.com/ilovephysics

Suppose you're on a game show, and you're given the choice of three doors. Behind one door is a car, behind the others, goats. You pick a door, say #1, and the host, who knows what's behind the doors, opens another door, say #3, which has a goat. He says to you, 'Do you want to pick door #2?' Is it to your advantage to switch your choice of doors?

Following with Python

Download the notebook from the canvas page for this week's material:
[Funcs.py](#)

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marimo edit Funcs.py
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Probability Functions

- Probability Mass & Density Functions
- Parameters
- Cumulative Distributions
- Joint PDF
- Likelihood
- Transforms, Jacobian

Probability Distributions

A Probability Distribution Function is a function of one or more RVs and one or more parameters, and represents the probability of all the ranges of values the RV can take.

If the RV is:

Discrete: the distribution is a **Probability Mass Function**: a set of points.

Continuous: the distribution is a **Probability Density Function**: a smooth curve.

Note that Probability Distribution Functions are very often referred to as PDFs. Overloaded term also used for Probability Density Function.

Discrete versus Continuous: Examples

Discrete RVs

Continuous RVs

Note that measured data is always discrete, no matter the nature of the underlying RV.

Probability Mass Functions

A Probability Mass Function (**PMF**) is p_X where $X = X_{(1)}$ is a single discrete RV.

The **Support** S of a PMF (also PDF) is the set of all values with a non-zero probability of occurring.

A PMF must satisfy three conditions:

$$1 \quad p_X(x) > 0 \quad \forall \quad x \in S$$

$$2 \quad \sum_{x \in S} p_X(x) = 1$$

$$3 \quad P(x \in A) = \sum_{x \in A} p_X(x)$$

Discrete PMF Example: max score from two dice

Sample space is $S = \{(i, j) | i, j = 1, 2, 3, 4, 5, 6\}$: there are $i \cdot j = 36$ possible outcomes with equal probability.

The total probability is 1, and is the sum of the individual probabilities, so

$$p_{ij} = \frac{1}{36}.$$

Some outcomes will appear more than once.

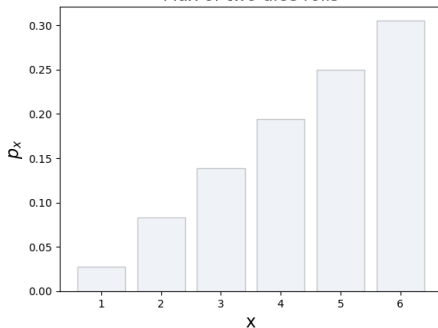
i, j	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	2	3	4	5	6
3	3	3	3	4	5	6
4	4	4	4	4	5	6
5	5	5	5	5	5	6
6	6	6	6	6	6	6

X	1	2	3	4	5	6
p_X	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

Discrete PMF Example: max score from two dice

X	1	2	3	4	5	6
p _x	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

Max of two dice rolls



Probability Density Functions

A Probability Density Function (**PDF**) is f_X where $X = X_{(1)}$ is a single continuous RV.

A PDF must satisfy the same conditions as a PMF, but we write the second and third in terms of integrals:

$$1 \quad f_X(x) > 0 \quad \forall \quad x \in S$$

$$2 \quad \int_S f_X(x) dx = 1$$

$$3 \quad P(x \in A) = \int_A f_X(x) dx$$

Probability: Area under distribution

Note that for a continuous RV X , the value of the PDF at some value of $X = x$ is not the probability of that value. The absolute y – axis value of a PDF has no meaning.

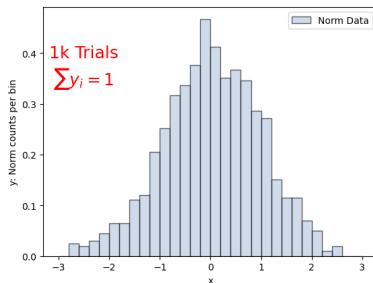
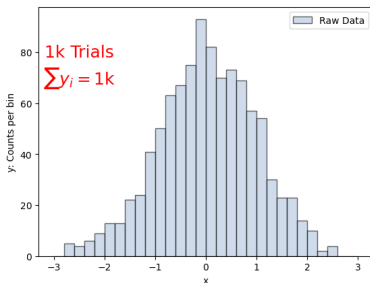
Recall that $\int_{-\infty}^{\infty} f_X dx = 1$, so f_X at any of the infinite possible values must be infinitesimally small: zero.

Trying to extract the probability of any exact value from a continuous PDF will return zero.

For continuous RVs, we can only talk about the probability of a value in a given range.

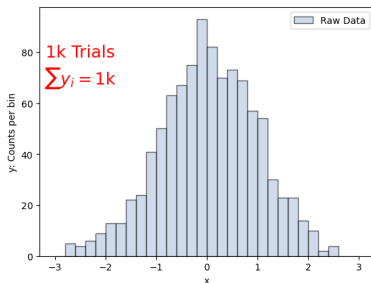
The Total Probability $\int_S f_X(x) dx = 1$

Spot the difference:

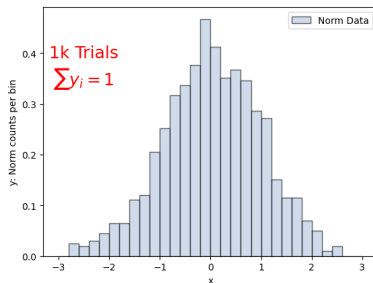


The Total Probability $\int_S f_X(x) dx = 1$

This is a regular distribution



This is a Probability Distribution



A PDF/PMF must be Normalised to unity, otherwise it does not satisfy the second axiom.

Reminder: Normalisation

Any function of a continuous RV can be turned into a PDF. For example:

$f(X; \theta) = x^3$ **is a function**

1 Write $f(X; \theta) = Ax^3$

2 Specify the 'range', for example $X : [0, 1]$

3 Set integral to one: $\int_0^1 Ax^3 dx = 1$

4 Solve for A: $A \frac{1}{4} = 1 \therefore A = 4$

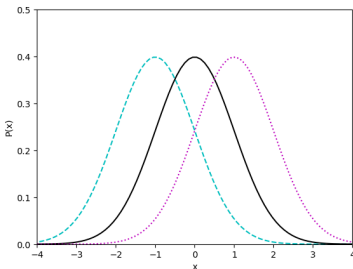
$f_X = 4x^3$ **for $0 \leq x \leq 1$ is a valid PDF**

Parameters

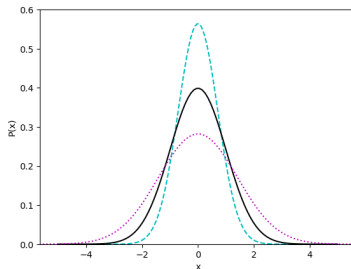
The **Parameters** θ of a PDF or PMF are classified by how they modify the behaviour of the function wrt the RV(s)

Consider a curve of some sort (a distribution function). If you move it to the left or right, it is still the same curve: has it shifted, or has the x-axis?

How about if you squish it or stretch it a bit?



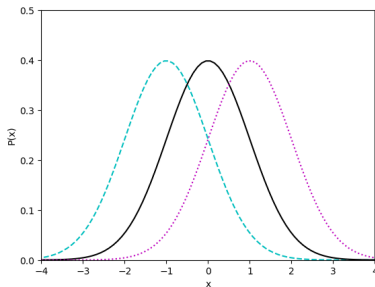
Lily



Funcs

Location Parameter

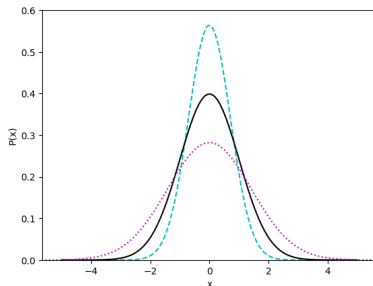
The **Location Parameter** of a PDF or PMF sets its position on the x-axis*.



For the Gaussian PDF in this example, the location parameter is μ , the True Mean.

Scale Parameter

The **Scale Parameter** of a PDF or PMF sets its spread on the x -axis*.



For the Gaussian PDF in this example, the scale parameter is σ , the Standard Deviation.

*Simplification Warning

Consider a PDF $f_X(X; \theta) = \theta_s(x - \theta_L)$.

The two parameters are $\theta = \{\theta_s, \theta_L\}$. The Scale Parameter is θ_s and the location parameter is θ_L

The inverse of a **Scale Parameter** is called a **Rate Parameter**

Note that the shifting and squishing/stretching analogy above is a pedagogic simplification.

Distribution functions often have forms like $f_X \propto \lambda^x$ (cf Poisson), where the parameter λ is a Rate Parameter (which we defined as squish/stretch) that also shifts the distribution peak left and right (which in our oversimplification we claimed was the job of the Location Parameter).

Normalisation versus Scale

We saw that $f_X = 4x^3$ is a valid PDF because it has integral 1.

My simplified explanation of the scale parameter might lead you to believe that using it could ruin the validity of the PDF.

Scale and Location parameters cannot ‘break’ the normalisation of a PDF.

The normalisation is not optional. It is fixed. If we change the scale/location, the range of validity (the domain) must also change.

Shape Parameter, Examples

The **Shape Parameter** alters the curve more profoundly, causing the peak to move while also affecting the symmetry of the distribution.

More on this: The Gaussian does not have a shape parameter, but there are PDFs that do (Beta, T, ..): please enjoy this fantastic interactive page: [Probability Playground](#) . It allows you to change the parameters of pretty much any distribution you can think of.

Scipy: The scale and loc parameters are used in the [scipy stats library](#), which has an extensive range of probability distributions. The scipy stats distribution functions can also take parameters as arguments which are not strictly classified as scale or loc. An example of a distribution like this is the Binomial, which we will cover soon. Its parameters do shift and squish/stretch the distribution, but they don't *only* shift and stretch/squish the distribution.

Cumulative Distribution Function

A **CDF** (Cumulative Distribution Function) F_X allows us to calculate the probability of X being measured as **any value up to and including a given value**.

$$F_X(x) = P(X \leq x) = \sum_{x_i \leq x} p_X(x_i) : \text{CDF for Discrete } X$$

Example: X is the maximum value from rolling two dice

X	1	2	3	4	5	6
p_X	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$
F_X	$\frac{1}{36}$	$\frac{4}{36}$	$\frac{9}{36}$	$\frac{16}{36}$	$\frac{25}{36}$	$\frac{36}{36}$

$$F_X(4) = P(X \leq 4) = \frac{16}{36}$$

CDF: Continuous RV

A **CDF** (Cumulative Distribution Function) F_X allows us to calculate the probability of X being measured as **any value up to and including a given value**.

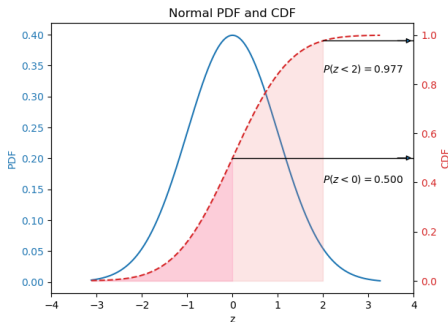
$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(x) dx : \text{CDF for Continuous } X$$

Recall: **For continuous RVs, we can only talk about the probability of a value in a given range.**

PDF and CDF

f_X : PDF gives us a shape.

$F_X = \int_{-\infty}^x f_X(x) dx = P(X \leq x)$: CDF gives us a Probability



Properly defining the PDF

The PDF we made earlier is $f_X = 4x^3$ for $0 \leq x \leq 1$.

To properly define this PDF, we need to also specify what it does outside the range $0 \leq x \leq 1$:

$$f_X(x) = \begin{cases} 0 & x < 0 \\ 4x^3 & 0 \leq x \leq 1 \\ 0 & x > 1 \end{cases}$$

Properly defining the continuous CDF

The CDF is $F_X(x) = \int_{-\infty}^x f_X(x) dx$

For this $f_X = 4x^3$ we have three regions.

$$F_X(x) = \int_{x < 0} f_X(x) dx + \int_0^1 f_X(x) dx + \int_{x > 1} f_X(x) dx$$

$$F_X(x) = \begin{cases} 0 & x < 0 \\ x^4 & 0 \leq x \leq 1 \\ 0 & x > 1 \end{cases}$$

Probability for a CRV

$$F_X(x) = \begin{cases} 0 & x < 0 \\ x^4 & 0 \leq x \leq 1 \\ 0 & x > 1 \end{cases}$$

The Probability of $X \leq 0.1$:

$$F_X(0.1) = \int_{-\infty}^0 4x^3 dx + \int_0^{0.1} 4x^3 dx = 0 + x^4 \Big|_0^{0.1} = 0.1^4 - 0 = 0.0001$$

Probability for a CRV

$$F_X(x) = \begin{cases} 0 & x < 0 \\ x^4 & 0 \leq x \leq 1 \\ 0 & x > 1 \end{cases}$$

The Probability of $0.45 < X \leq 0.55$:

$$F_X(0.45) = \int_0^{0.45} 4x^3 dx = x^4 \Big|_0^{0.45} = 0.45^4 - 0$$

$$F_X(0.55) = \int_0^{0.55} 4x^3 dx = x^4 \Big|_0^{0.55} = 0.55^4 - 0$$

$$F_X(0.55) - F_X(0.45) = 0.55^4 - 0.45^4 = 0.0505$$

Probability for a CRV

$$F_X(x) = \begin{cases} 0 & x < 0 \\ x^4 & 0 \leq x \leq 1 \\ 0 & x > 1 \end{cases}$$

The Probability of $X > 0.95$:

$$F_X(0.95) = \int_{0.95}^1 4x^3 dx = x^4 \Big|_{0.95}^1 = 1 - 0.95^4 = 0.1855$$

Joint Probability

The **Joint CDF** for two RVs Y, Z allows us to calculate the joint probability of the two RV measurements.

$$P(\{Y = y\} \cap \{Z = z\}) = F_{YZ}(y, z; \theta)$$

A Joint PDF must satisfy:

1 Normalisation: $0 \leq f_{YZ} \leq 1$

2 Total probability must be 1: $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{YZ} dy dz = 1$

Note that $Y = y$ is shorthand for $(y - \delta) \leq Y \leq (y + \delta)$

Joint Probability for two Independent RVs

$$P(\{Y = y\} \cap \{Z = z\}) = F_{YZ}(y, z; \theta)$$

Recall the Conditional Probability: $P(A \cap B) = P(A|B)P(B)$

If the RVs are Independent, this is $P(A \cap B) = P(A)P(B)$

In that case, we can write the **Joint PDF for Independent RVs**:

$$f_{YZ}(y, z; \theta) = f_y(y; \theta) f_z(z; \theta)$$

The independence allows us to factorise the joint PDF into a product of the PDFs for the individual RVs.

Independent RVs

Two random variables Y and Z are independent if knowing the value of one of them does not change the probabilities for the other one.

Are these pairs **Independent**, **Dependent**, or **Need to look into that**

- Y : coin toss, Z : another coin toss
- Y : speed of a car, Z : kinetic energy of a car
- Y : my assigned sex, Z : my IQ

Joint PDF: Example

Suppose Y and Z have values in $[0,1]$ with $f_{YZ}(y, z) = 4yz$.

Check f_{YZ} is a valid:

$$\int_0^1 \int_0^1 4yz \, dy \, dz = \int_0^1 [2y^2z]_0^1 \, dz = \int_0^1 [2z] \, dz = [z^2]_0^1 = 1$$

Calculate the joint probability $P(Y \leq 0.2 \cap Z > 0.3)$:

$$\int_{0.3}^1 \int_0^{0.2} 4yz \, dy \, dz = \int_{0.3}^1 [2y^2z]_0^{0.2} \, dz = \int_{0.3}^1 [0.08z] \, dz = [0.04z^2]_{0.3}^1 = 0.0364$$

Marginal PDF

We can extract the **Marginal PDF** for each of the RVs by integrating our joint PDF over the other RV:

$$f_Y(y; \theta) = \int_{-\infty}^{\infty} f_{YZ}(y, z; \theta) dz$$

Similarly for Z:

$$f_Z(z; \theta) = \int_{-\infty}^{\infty} f_{YZ}(y, z; \theta) dy$$

This is very handy as it allows us to look at one RV at a time.

Marginal PDF: Example

$$f_{YZ}(y, z) = 4yz.$$

$$f_Y(y) = \int_0^1 4yz \, dz = [2y z^2]_0^1 = 2y$$

$$f_Z(z) = \int_0^1 4yz \, dy = [2y^2 z]_0^1 = 2z$$

Are Y and Z independent?

More than two RVs

In the most general case of Z RVs, the notation becomes unsavoury.

Joint PDF for independent RVs:

$$f_X = \prod_j^Z f_{X_{(j)}}$$

Marginal PDF for $X_{(1)}$ is:

$$f_{X_{(1)}} = \int \cdots \int f_X dx_{(2)} dx_{(3)}, \dots, dx_{(Z)}$$

Joint CDF is:

$$F_X = \int \cdots \int f_X dx_{(1)} dx_{(2)}, \dots, dx_{(Z)}$$

Likelihood: not a Joint PDF

The **Likelihood** for a single RV Y is :

$$\mathcal{L}(Y; \theta) = \prod_i^N f_Y(y_i; \theta_i)$$

Compare with the joint PDF for Z RVs:

$$f_X(x; \theta) = \prod_j^Z f_{X_{(j)}}(x_{(j)}; \theta_j)$$

These look similar but are not!

The **Likelihood** is a product over the N **Measured Values** of the PDF.

Likelihood: not a Probability

The Likelihood is the product of PDFs (or PMFs), so cannot adhere to the Normalisation Axiom itself, but we can **Normalise** it.

The Likelihood of θ given the data:

$$\mathcal{L}(y; \theta) = \prod_i^N f_Y(y_i; \theta_i)$$

The **Normalised Likelihood** is the Probability of **The Data** given θ

$$P(y|\theta) = \frac{\mathcal{L}(y; \theta)}{\int \mathcal{L}(y; \theta) d\theta}$$

If this feels like it should be the other way around to you, it may comfort you to know that I got it the other way around, but this is the right way!

Likelihood: features in Bayes' Theorem

The Likelihood is for θ , given the data.

The **Normalised Likelihood** is the Probability of **The Data** given θ

$$P(y|\theta) = \frac{\mathcal{L}(y; \theta)}{\int \mathcal{L}(y; \theta) d\theta}$$

Recall Bayes Theorem:

Bayesian

Frequentist

$$P(H_i|X) = \frac{P(X|H_i)\pi(H_i)}{\sum_i P(X|H_i)\pi(H_i)} \rightarrow \frac{P(x|\theta)\pi(\theta)}{\int P(x|\theta)\pi(\theta)d\theta} \rightarrow \frac{P(x|\theta)}{\int P(x|\theta)d\theta} = P(\theta|x)$$

The Normalised Likelihood is the Relative Probability of our Hypothesis, given the data. Frequentists do not accept that priors on hypotheses $\pi(\theta)$ have any place in probability calculations. They rely entirely on the Likelihood.

Likelihood of the parameter, given the data

The Likelihood is telling us something entirely different from the PDF.

PDF: Fixed parameter θ provides the shape/scale/location of the f_X . We plot the PDF against the (possible measurements of) x .

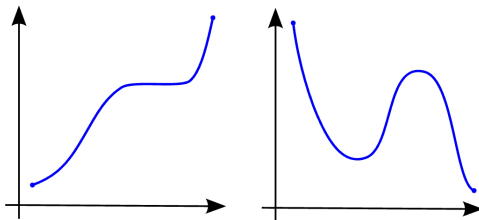
Likelihood: Measured data x provide a histogram which gives us a clue as to the form (parameters) of the f_X . We plot the Likelihood against the (possible values of) the unknown parameter θ .

We will talk about Likelihood loads more. This is just a taster.

Changing Variable

In real life we often cannot measure the RV we want directly.

We measure X and want the PDF for $Y = f(X)$. What do we do? It depends on the relationship $Y = f(X)$: is it monotonic (left pic) or not (right pic).



By Oleg Alexandrov

Changing Variable $Y = f(X)$: 1D

Assume that Y is a **monotonically increasing** function of X . This just means that Y increase if X increases, and doesn't have turning points.

$$Y = g(X)$$

$X = g^{-1}(Y)$ is the inverse relationship.

We can therefore write:

$$P(Y \leq y) = P(g(X) \leq y) = P(X \leq g^{-1}(y)) = P(X \leq x)$$

Where $y = g(x)$, so $g^{-1}(y) = g^{-1}(g(x)) = x$.

$$P(Y \leq y) = P(X \leq x)$$

Changing Variable $Y = f(X)$: 1D

We have shown that: $P(Y \leq y) = P(X \leq x)$

Therefore the **CDF for Y** is:

$$F_Y(y) = F_X(x)$$

Therefore the **PDF for Y** is:

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} F_X(x) = \frac{dx}{dy} \frac{d}{dx} F_X(x) = \frac{dx}{dy} f_X(x)$$

$$f_Y(y) = \frac{dx}{dy} f_X(x)$$

Changing Variable $Y = f(X)$: 1D

$f_Y(y) = \frac{dx}{dy} f_X(x)$: for a monotonically increasing function

Generally (monotonic increasing or decreasing):

$$f_Y(y) = \left| \frac{dx}{dy} \right| f_X(x)$$

If the function is not monotonic, we must 'divide and conquer': split the function up into ranges over which it is monotonic, and do the transform piece-by piece. See eg [this blog](#) for an example. You don't need this for this module.

Multivariate Transform

The PDFs $f_X(x)$ and $g_Y(y)$ are multivariate, with $X = \{X_{(1)}, X_{(2)}\}$ and similarly for $Y = \{Y_{(1)}, Y_{(2)}\}$.

The $X_{(j)}$ are again related to the $Y_{(j)}$, but we now have $Z^2 = 4$ relationships. Matrices are our friend here.

$$\begin{bmatrix} f_{Y_{(1)}} \\ f_{Y_{(2)}} \end{bmatrix} = \begin{bmatrix} \frac{\partial x_{(1)}}{\partial y_{(1)}} & \frac{\partial x_{(1)}}{\partial y_{(2)}} \\ \frac{\partial x_{(2)}}{\partial y_{(1)}} & \frac{\partial x_{(2)}}{\partial y_{(2)}} \end{bmatrix} \begin{bmatrix} f_{X_{(1)}} \\ f_{X_{(2)}} \end{bmatrix}$$

Equivalently, we can write $f_Y = \frac{1}{J} f_X$ where J is the **Jacobian of Y** .

The Jacobian

The **Jacobian of Y** is the **Determinant** of the Z^2 matrix of partial derivatives $\frac{\partial y_{(j)}}{\partial x_{(j)}}$.

This allows us to make transforms between variables in multivariate PDFs.

$$f_Y = \frac{1}{J} f_X$$

$$J = \begin{vmatrix} \frac{\partial y_{(1)}}{\partial x_{(1)}} & \cdots & \frac{\partial y_{(1)}}{\partial x_{(Z)}} \\ \vdots & \ddots & \vdots \\ \frac{\partial y_{(Z)}}{\partial x_{(1)}} & \cdots & \frac{\partial y_{(Z)}}{\partial x_{(Z)}} \end{vmatrix}$$

Jacobian: Example

Suppose points are randomly distributed in a Cartesian (x, y) plane. How are they distributed in polar coordinates (r, θ) ?

Relationships:

$$r = (x^2 + y^2)^{1/2}, \quad \theta = \arctan \frac{y}{x}$$

Partial Derivatives:

$$\frac{\partial r}{\partial x} = \frac{1}{2}(x^2 + y^2)^{-\frac{1}{2}} \cdot 2x$$

$$\frac{\partial r}{\partial y} = \frac{1}{2}(x^2 + y^2)^{-\frac{1}{2}} \cdot 2y$$

$$\frac{\partial \theta}{\partial x} = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \left(-\frac{y}{x}\right)$$

$$\frac{\partial \theta}{\partial y} = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \frac{1}{x}$$

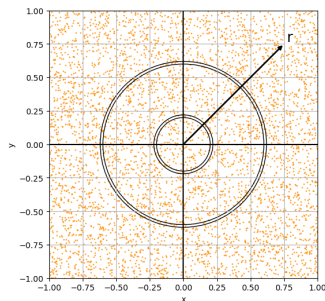
Jacobian: Example cont.

Jacobian:

$$J = \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{vmatrix} = \frac{\partial r}{\partial x} \cdot \frac{\partial \theta}{\partial y} - \frac{\partial r}{\partial y} \cdot \frac{\partial \theta}{\partial x} = \frac{1}{r}$$

Think:

$f(r, \theta) = r \cdot g(x, y)$: this tells us that if $g(x, y)$ is uniform, then $f(r, \theta)$ is linearly increasing with r .



Thanks For Your Time

Fin.